

(B.pharm, MSc. in Health Supply Chain Management)

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Course Syllabus



Module Name: Social and Administrative Pharmacy



Module Module Code: Phar-M4211



Course code: Phar 4212



Course title: Pharmaceutical Supply Chain Management



Course ECTS credits: 7 ECTS



2025/2026 Content covered

<u>Selection of pharmaceuticals</u>	<u>04</u>
<u>Quantification</u>	<u>05</u>
<u>Procurement of pharmaceuticals</u>	<u>06</u>
<u>Inventory management</u>	<u>07</u>
<u>Inventory control system</u>	<u>08</u>
<u>Logistics management information system</u>	<u>09</u>
<u>Stock status assessment</u>	<u>10</u>
<u>Managing for rational use medicines</u>	<u>11</u>
<u>Monitoring and evaluation</u>	<u>12</u>

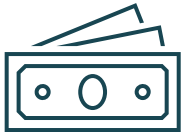
Why worry about Pharmaceuticals and their management?



Pharmaceuticals save lives and improve health



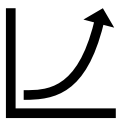
Promote trust and participation in health services.



Costly for individuals, households, government health institutions, country.



Are different from other consumer products.



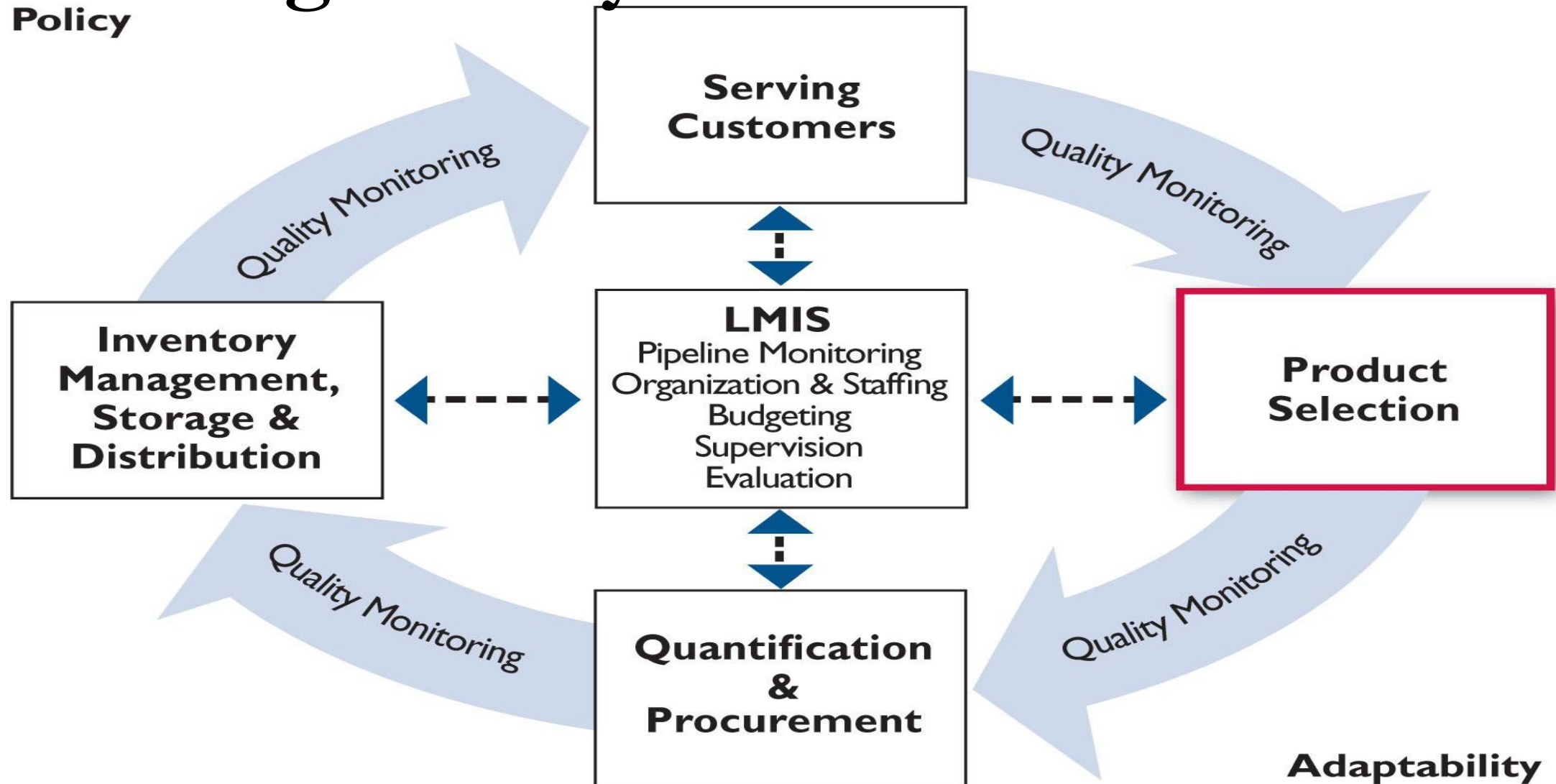
Substantive improvements in the supply and use of drugs are possible.

Overview

- Managing drug supply is organized around the following basic functions:
 - *Selection*
 - *Quantification*
 - *Procurement*
 - *Distribution and*
 - *Use*

The logistics cycle

Policy



Chapter 4: Selection of pharmaceuticals

Selection

- Product selection refers to the process by which health programs, as a whole, **select, evaluate and ultimately procure the products** that will be used and consumed in service delivery.
- It is directly linked to *serving customers* by defining **what products are procured and used** in the health system and the range of products that a customer can receive

Cont . . .

- Standardization
 - Enables programs to make decisions by question and the avoidance or proliferation of similar products.
- Limiting the variety of products that are used and available at public sector facilities can make the supply chain more manageable.
- Fewer products enhance the **agility, manageability, and efficiency** of a supply chain.

Cont . . .

- The product selection process is informed by **local policies and guidelines** as well as **procedure guides and protocols**.
- Pharmaceutical products are selected from or become part of a national essential medicines list (**EML**) and are based on standard treatment guidelines (**STGs**);
- Products **must be registered** for overall use in-country.

- The rationale for selecting a limited essential health product is that it may lead to **better supply, more rational use and lower cost.**
- It done as:
 - Health products represent **a large part of the health program budget but funds** are limited
 - **Large numbers of drugs are available**, but it is impossible to keep up-to date with all the drugs in the market

Basic steps in drug selection

- Establish **Drug Selection Committee**
- Determine the **prevalent health problems and patient characteristics**
- Decide which **health problems may be treated at the level of drug selection**
- Choose **the drugs to be used** for the health problems
- **Structure the list**
- **Introducing the list of drugs to health professionals** of institution and other concerned bodies
- **Updating** the list of drugs

- The WHO criteria for selection of essential medicines
 - ❑ Pattern of prevalent disease
 - ❑ Treatment facilities
 - ❑ Training and experience of available personnel
 - ❑ Financial resources
 - ❑ Genetic, demographic and environmental factor
 - ❑ Medicines which have sound and adequate data on **efficacy** and **safety** (clinical studies, & evidence from general setting)
 - ❑ Adequate quality including bioavailability and stability under anticipated conditions of storage

- When two or more medicines appear to be similar in the above respects
 - The choice is made by carefully evaluating their ***relative efficacy, safety, quality, price and availability.***

– Cost

- Consider the cost of the **total treatment** rather than the **unit cost** of the medicine
- The basis of a cost-effectiveness analysis.

– Other factors such as

- Pharmacokinetic properties and availability of facilities for storage.

– Essential medicines formulated as single compounds.

- But fixed-ratio combination products are acceptable only when
 - The dosage of each ingredient meets the requirements of a defined population
 - The combination has a proven advantage over single compounds
 - Administered separately in therapeutic effect, safety or compliance

Roles of health supply chain personnel

- To ensure that products selected in the public sector health system are included on the national EML
- To update regularly to address any new products on the market
- To collect products consumption data & convert to appropriate stock keeping units

Advantages of a limited list of pharmaceuticals

Major objective	Addressed Challenge
Supply	Easier procurement, storage, and distribution Lower stocks Better quality assurance Easier dispensing
Cost	Lower prices More competition
Patient use	Focused education efforts Reduced confusion and increased adherence to treatment Improved drug availability
Health care provider	Focused and up-to-date information More experience with fewer medicines Reduction of AMR Better recognition of ADR

Steps in the process of pharmaceuticals selection

1. Establish DTC or drug selection committee
2. Determine prevalent health problems based on the data
3. Select pharmaceuticals for each health problem following recent STG, National NCD protocol, EML; generic name, dosage form (s), strength(s)
4. Structure list of pharmaceuticals
5. Introduce the list for staffs
6. Update the list

5. Pharmaceutical Quantification

QUANTIFICATION

- Quantification is the first step in the procurement process.
- Quantification is the process used to determine how much of a product is required for the purpose of procurement.
- More specifically, quantification involves estimating not only the quantities needed of a specific items but also the financial means required for purchasing the items.

Cont. . .

- Needs are estimated for a given context, so the analysis must include contextual factors, such as available funds, human resources capacity, storage space capacity and capacity to deliver services.
- Quantification of health commodities follows a step-by-step approach.

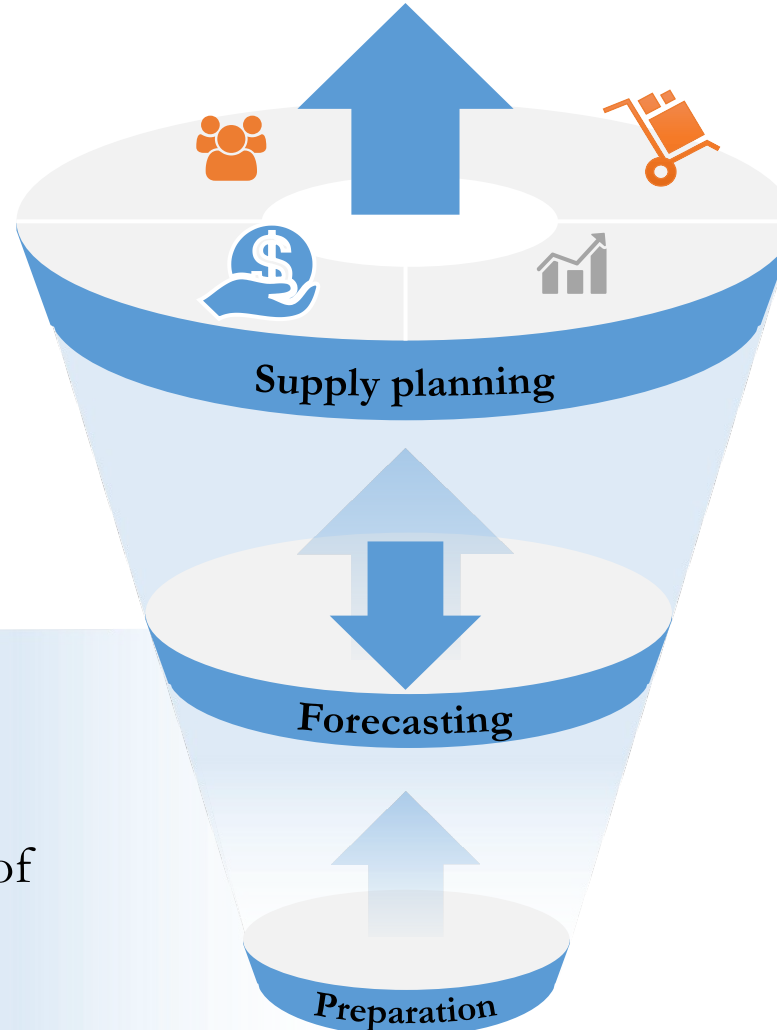
STEPS IN QUANTIFICATION

Supply Planning

- Organize, analyze, and adjust data
- Build supply planning assumptions
- Calculate total commodity requirements and costs
 - Develop supply plan
 - Compare costs to available funding

Preparation

- Describe the program
- Define scope and purpose of the quantification
 - Collect required data



Forecasting

- Organize, analyze, and adjust data
- Build forecasting assumptions
- Calculate forecasted consumption for each product
- Reconcile forecasts to produce a final estimate for each product

Cont. . .

Preparation

1. Describe the program

- ✓ Background information on the program should be collected
 - ✓ A review of the program goals, strategies, priorities, and any expansion plans or change in policies that will influence uptake of services and demand for commodities.
 - ✓ A brief description of the service delivery model, supply chain, level of political commitment, and financial support for services and for commodities.

Cont. . .

2. Define the scope and purpose of the quantification

- ✓ Scope of the program
 - (public, private and/ faith based; geographical region or specific population group; funding agency or implementing partner)
 - List of commodities.
- ✓ Identify the purpose and how it will address the program's needs.
- ✓ Define the time period for the quantification exercise which could range from two to five years.

3. Collect required data

This involves collecting data for forecasting and supply planning steps.

Data for forecasting include: consumption data, services data on the number of health services being provided, morbidity data, demographic data, and information on current program performance, expansion plans, strategies, priorities, and program targets for each year of the quantification.

Data for supply planning include National/program level stock on hand, expiration dates for products in stock, quantity on order, established shipment schedules and current shipment delivery schedule, established national/program level maximum and minimum stock levels, product information, supplier information, funding information, procurement information, and distribution information.

Forecasting

1. Organize, analyze, and adjust data
2. Build forecasting assumptions
 - Assumptions are made to adjust historical program data when it is of poor quality (incomplete, outdated, unreliable, or unavailable) and also for future program performance.

Cont. . .

- Assumptions may include issues such as:
 - Expected uptake in services
 - Compliance with recommended treatment guidelines
 - Future changes in standard treatment guidelines and/or introduction of new commodities
 - Impact of changing program policies and strategies on supply and demand
 - Service capacity (infrastructure, human resources availability, and capacity)
 - Client awareness of and access to services
 - Timing and amount of funding commitments for procurement
 - Seasonality
 - Geographic variations in disease incidence and prevalence
 - Other factors that might affect demand

Cont. . .

3. Calculate forecasted consumption for each product

The four general methods for quantification are:

- Consumption method
- Morbidity method
- Proxy consumption/adjusted consumption method
- Service level projection of budget requirements

Cont. . .

Selection of the appropriate quantification method requires considering:

- Advantages and disadvantages
- The conditions under which each method can be applied
- The sets of data required
- The organization of the drug supply system
- Additional purposes

A. Consumption method

- Involves using records of past consumption on individual medicines
- It can be used under the following conditions:
 - Adequate fund
 - Acceptable pattern of prescribing
 - Adequate and uninterrupted drug supply
 - Good stock management
 - Low level of wastages and losses

Cont. . .

Data sets needed:

- Consumption data
- Drug list
- Pharmaceutical budget

Cont. . .

Steps in quantification:

1. Prepare/review a list of drugs with order pack size and price data
 2. Select the period
 3. Enter consumption data for each drug
 - The total quantity used during the review period
 - The number of the days in which the drug was out of stock
 - The lead time for the last procurement (average from last several procurements)
- Sources of data: Stock records, distribution report, invoices from supplier, and dispensing records

4. Calculate average monthly consumption

$$C_A = C_T / (R_M - (D_{os} / 30.5)) \dots \dots \dots \text{equation 1 (Preferred)}$$

or

$$C_A = C_T / (R_M - M_{os}) \dots \dots \dots \text{(Alternative)}$$

Where,

C_A = Average monthly consumption, adjusted for stock out

C_T = Total consumption during the review period in in basic unit

R_M = Review period in months

D_{os} = Number of days an item was out of stock during the review period

5. Calculate projected AMC for **expected** changes in consumption pattern

$$C_P = C_A + (C_A \times A_U)$$

Where C_P : Projected average monthly consumption

A_U : Use adjustment

6. Calculate the safety stock needed for each drug

$$\mathbf{SS = C_p \times L_T} \dots\dots\dots \text{equation 2}$$

Where,

SS = Safety stock

$\mathbf{C_p}$ = Projected average monthly consumption

L_T = Average lead time

7. Calculate the quantity of each drug required in the next procurement period

$$\mathbf{Q_o = C_p \times (L_T + PP) + SS - (S_i + S_o)}$$

Where, QO = Quantity to order before adjustment for losses or program change

PP= Procurement period

S_i = Stock now in inventory

So= Stock on order

8. Adjust for losses

$$QA=Qo+(Qo \times AL)$$

9. Compile decentralized quantification (if applicable)

10. Estimate **costs** for each drug and **total** costs

11. Compare total cost with budget and make adjustment

–Either obtain **more funds** or

–**Reduce** the number of drugs/or quantities ordered.

Exercise I

- Calculate the amount of Metronidazole 250 mg capsule for the next six months (procurement period) based on the following data
 - $C_T = 89,000$ capsule
 - $S_o = 58,000$ caps
 - $D_{os} = 34$ days
 - Loss/adjustment for 6 month = 10%
 - Estimated purchase price per capsule = 2.50 ETB
 - A box contains 500 capsules
 - $S_i = 81,000$ caps
 - $L_T = 3$ months
 - Use is expected to ↑ by 5%

Cont. . .

Advantages:

- First choice for quantification, given reliable data
- Requires less detailed calculation
- Useful for facilities where health problems are numerous and drug treatment is complex
- Identifies stock management problems and encourages improvement
- No morbidity data and treatment schedule needed

Cont. . .

Disadvantages:

- Reliable consumption data may not be found
- Can perpetuate irrational use
- May not be helpful for reviewing prescribing/drug use
- Less appropriate if there have been long stockouts/ high drug wastages or loses
- Doesn't encourage good morbidity recording

B. Morbidity method

- Estimates the need for specific medicines based on:
 - expected number of attendances,
 - the incidence of common diseases and
 - standard treatment patterns for the diseases considered.
- **Assumption:** service providers will adhere to established treatment guidelines.

Cont . . .

- Conditions:
 - Available consumption data are incomplete or unreliable
 - Prescribing patterns are inappropriate
 - The budget is insufficient
 - The health facilities or services concerned are new, or expanding or contracting rapidly.

Cont . . .

- Data sets needed:
 - Drug list with pack size and price data
 - Complete patient morbidity profile
 - Standard drug treatment schedules
 - Pharmaceutical budget

➤ Steps

1. Specify the list of health problems
2. Establish the list of medicines to be quantified
3. Establish standard or average treatments

– $Q_E = D_{CU} \times N_D \times L_D$

– Q_E = Quantity of medicine needed for each treatment episode

N.B: Treatment episode is a patient contact for which a standard course of drug treatment is required

– D_{CU} = The number of basic unit in each average doses for health problem in question

– N_D = Average number of doses of each medicine per day for the problem

– L_D = Average number days of treatment for each medicine per episode

4) Collect morbidity data for each health problems treated

- $C_E = C + (C \times A_U)$

- C_E = Expected total number of contacts in thousands

- C = Last total number of contacts

- A_U = Expected rate of increase of contact

- Sources of information for morbidity data

- Epidemiological surveillance data

- Research data on incidence and prevalence of disease

- Health conditions in a given population

5) Calculate the number of *treatment episodes* for each health problem

$$-E_T = C_E \times F$$

– E_T : Number of treatment episodes

– C_E : Expected total number of contacts

– F : Number of treatment episodes per 1,000 contacts

• This calculation must be done separately for each discrete age range

6) Estimate the quantity of drugs needed for each health problem

$$-Q_T = E_T \times Q_E \times P_T$$

– Q_T : Quantity of drugs needed

– E_T : Number of treatment episodes

– Q_E : Quantity of drugs for each treatment episode

– P_T : Percentage of cases that are treated with that specific regimen

- 7) Combine the estimates for each drug from the various health problems into a master procurement list
- 7) Adjust quantities to cover other health problems for filling the pipeline and current stock position, and for expected losses
- 7) Estimate costs for each drug and total costs
- 7) Compare total costs with budget and make adjustment

Exercise II

- Estimate the total quantity of medicines required for the *treatment of otitis media* based on the following information.
 - Last total number of contacts (C) = 3,123,408
 - 5% expected rate of increase
 - Frequency treatment episodes per 1000 contacts (106 for <5, 29 for >5)

- Treatment for otitis media
- Under 5
 - Cotrimoxazole suspension 5ml BID for 10 days
 - Paracetamol solution of 120mg/5ml, 1.5ml QID for 10 days
 - Pseudoephedrine syrup 2.5ml QID for 5 days

- Above 5
 - Cotrimoxazole 800/160 mg tab BID for 10 days
 - Paracetamol 500 mg tab, 2 tab QID for 5 days
 - Pseudoephedrine 60 mg tab QID for 5 days
- All patients of otitis media are expected to be treated by cotrimoxazole and paracetamol, whereas 80% of patients treated with pseudoephedrine

Cont . . .

Advantages:

- Does not require consumption data
- Can be used for review of drug use and prescribing
- Can be used for estimating need in new and scaling-up programs or disaster assistance
- Motivates reliable morbidity recording

Cont . . .

Disadvantages:

- Detailed morbidity data and standard treatment schedule may both present difficulty
- Require detailed calculation
- Standard treatment may not really be used

C. Proxy/adjusted consumption method

- Used when supply systems face a severe information deficit (consumption and morbidity data).
- Uses known consumption data from one system “standard” to estimate the drug needs in a similar or expanded system “target”
 - Population based: Defining drug use per 1,000 population
 - Service based: Defining drug use per specified patient case

Cont. . .

Steps in quantification:

1. Select the standard system for comparison and extrapolation
2. Develop the drug list
3. Establish the time period to be covered in review
4. Review records from the standard system to compile contact or population data
5. Establish the denominator for extrapolation

Cont. . .

6. Determine the consumption rate in the standard system

$$\text{Standard consumption rate} = \frac{\text{standard adjusted annual consumption}}{\text{number of thousands of contacts or inhabitants}}$$

7. Extrapolate the standard system's consumption rate to the target system

Target consumption = (Standard consumption rate) x (Number of thousands of contact or inhabitants)

Cont . . .

8. Adjust for expected losses
9. Estimate costs for each drug and total costs and make adjustments

D. Service-level projection of budget requirements

- It uses the average drug procurement cost per attendance or bed-day in different types of health facilities in a standard system to project drug costs in similar types of facilities in the target system.
- This method does not estimate quantities of individual medicines rather it is used to estimate financial requirements.

- Steps

- 1) Establish the categories of **facilities** and determine the number in each category **standard**

- 2) Determine the patient contact denominator for each type of facility and compile or estimate the average number of patients contacts of each type at each category of facility

- 3) Calculate the **average cost** per contact

- The **total drug purchases** for the facility or facilities in class divided by the attendances or **bed-days**

- 4) Calculate the total projected drug costs

$$\text{Average cost per contact} = \frac{\text{Total drug purchase}}{\text{Total attendance or bed days}}$$

Cont. . .

COMPARE AND RECONCILE RESULTS OF DIFFERENT FORECASTS

- If availability and quality of data permits, different data types should be used to generate multiple forecasts.
- Evaluate the quality of each data type (completeness, accuracy, timeliness, and availability)
- Reliability of intermediate variables: assess whether these variables are based on current and accurate local data. Variables include dispensing protocols, population of program covered or reached by program, assumptions, disease prevalence, population growth rate, adherence to STGs or testing protocols, and scale-up factor.

Supply planning

- The final decision on the quantities to procure is based on the amount of funding available for procurement of products.
- If sufficient funding is available, the final quantity to procure for each product will be the same as the quantity to order that was determined during the quantification.

Supply Planning. . .

- Organize and analyze data
 - Data for the supply planning step are different from the data for the forecasting step
 - However, you can collect data for both the forecasting and the supply planning steps at the same time
 - The supply planning data focuses on
 - Product, supplier, procurement, funding, distribution, and stock status

- Build supply planning assumptions
 - To account for missing or low-quality data
 - Timing of available funds
 - Amount of available fund
 - Lead times for each supplier
 - Minimum and maximum stock levels for each level in the system.

- Estimate total commodity requirements and costs
 - To determine the quantity of each product needed to meet the forecasted consumption(i.e. the output from forecasting)
 - To ensure the in-country supply pipeline has adequate stock levels to maintain a continuous supply to service SDPs
 - Costs per procurement unit plus freight, customs, and other shipping fees will give you a total cost.

- Develop the supply plan
 - The shipment quantities and delivery schedules to ensure a continuous supply of products to the country
 - Developing the supply plan helps program managers to
 - Track forecasted consumption data
 - Identify funders and funding commitments by product
 - Identify suppliers for each product
 - Coordinate timing of funding commitments and procurements
 - Schedule shipments according to lead times and stock levels in country
 - Stock outs and losses due to overstocking and expiry

- Compare costs to available funding
 - The final decision on the quantities to procure by the amount of funding available for procurement of products
 - Adjusting the forecasting assumptions will reduce the total quantities of products expected to be dispensed or used
 - Thereby, reducing the overall total commodity requirements and costs.

Using the quantification results

- The quantification team should formally present the results of the quantification to stakeholders.
- Presenting the results of the quantification is an opportunity for the team to describe the national stock status of commodities to all stakeholders
- These quantification outputs enable program managers, funders, buyers, and suppliers to plan and schedule their inputs.
- ❖ The quantification exercise is also an opportunity to identify and advocate for other supply chain improvements.

Indicators of poor quantification:

- Shortage of drugs
- Overstocking of drugs
- Irrational prescribing
- Inefficient budget use
- Suppression or distortion of demand

THANK YOU!!